Emergency Response Agent (ERA): Concept Note

Aligned SDGs: SDG 11 (Sustainable Cities & Communities), SDG 13 (Climate Action)

Category: Agentic AI System

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Project Title: AIDRAC:- Agentic AI Emergency Response Agent

1. Problem Statement

Natural disasters such as floods, cyclones, earthquakes, wildfires, and landslides cause significant damage to human life and infrastructure. During emergencies, people often struggle to access real-time information about safe evacuation routes, nearby shelters, hospitals, and emergency services.

2. Proposed Solution

The Emergency Response Agent (ERA) is an Agentic AI-powered system designed to assist citizens and authorities during disasters. The system autonomously collects and analyzes real-time weather, traffic, and location data to provide intelligent emergency support.

3. Key Features

• Real-time disaster monitoring
• Intelligent evacuation route planning
• Shelter and hospital recommendations
• Automated emergency alerts
• Multi-agent AI coordination
• Citizen and authority dashboard

4. Technology Stack

Frontend: React.js
Backend: FastAPI
Database: PostgreSQL
AI Framework: LangGraph / CrewAI
LLM: OpenAI GPT or Google Gemini

5. Expected Impact

• Faster emergency response
• Reduced loss of life and property
• Improved disaster preparedness
• Better coordination among emergency agencies

6. Target Users

• Citizens
• Disaster Management Authorities
• Municipal Corporations
• Emergency Response Teams
• NGOs

7. Conclusion

The Emergency Response Agent uses Agentic AI to provide proactive and intelligent disaster response, improving public safety and emergency coordination.